Image Captioning via Depth-Aware Scene Graphs and Spatio-Structural Modeling with LLM Fine-Tuning

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Abstract. This paper proposes a method for image captioning based on Depth-Aware and Commonsense-Augmented Scene Graphs (DASG-CS) integrated with a large pre-trained language model fine-tuning strategy. The proposed approach constructs structured contextual representations by combining object-level visual information, quantified three-dimensional spatial relationships, and semantic and commonsense knowledge from multiple sources. On the basis of these structured representations, a T5-based language generation model is fine-tuned to produce image captions with enhanced spatial awareness and contextual coherence, without requiring direct visual input to the language model. The overall architecture is implemented through three main stages: (1) depth-aware visual feature extraction, (2) multi-source knowledge graph construction and fusion, and (3) structured representation-driven caption generation. Experimental evaluations conducted on the MS-COCO and Flickr30k benchmark datasets demonstrate consistent improvements in BLEU-n metrics compared to representative baseline methods, particularly in accurately describing relative position, depth cues, and spatial structure between objects. These results highlight the effectiveness of integrating depth-aware spatial modeling and knowledge graphs into LLM-based image captioning frameworks.

Keywords: Image captioning · Knowledge graph · Depth-Aware Scene graph · LLM Fine-Tuning.

1 Introduction

Generating a caption for an image with high content and semantic accuracy using natural language words for each image is a challenge for image captioning in the field of computer vision, combined with natural language processing [5, 15]. To implement this problem, the system needs to combine many elements, including object recognition in the image and contextual analysis of the relationship between objects. In addition, the generated caption must also create coherent text, appropriate to the context, and accurately reflect the semantic meaning of the input image [15, 17], which is a major challenge. Recent works applying deep learning models, especially the application of Transformer-based models and large language models (LLMs) in visual-linguistic programming (VLP) training, have significantly improved the accuracy and time of extracting captions for images. However, a major limitation and challenge is that current VLP image description methods rely primarily on visual features and image semantics, which perform quite well, but cannot understand depth representation in space. To improve this problem, this paper proposes a method to represent depth-enhanced spatial relationships between objects in images, especially those with complex relationships and where relative spatial distances are difficult to distinguish.

To generate an accurate semantic caption, the model needs to be able to accurately identify objects, determine the relationships between them, and create a coherent and contextually appropriate description of the image [15, 17]. Recent advances in deep learning, especially Transformer models and large language models (LLMs) in visual-linguistic programming (VLP) training, have significantly improved the accuracy of generating captions that accurately reflect the content and semantics of images [7]. Most traditional caption generation models use only 2D features from RGB images and offer limited 3D spatial scene representation. As a result, these models struggle to describe location, depth, proximity, or spatial layering between objects. For instance, the caption "the car is closer to the building" lacks clear distance or depth cues, making spatial inference difficult. To address this, this paper introduces a method for extracting enhanced object-to-object spatial relationships based on depth, enriching spatial semantics and improving caption generation accuracy.

The main contributions of this work are: (1) Propose a DASG-CS structure, combining visual features, quantitative depth information, and common sense knowledge to represent the image scene uniformly. (2) Develop a spatially perceptive LLM refinement strategy, allowing the model to utilize spatial-structural relationships and semantic knowledge in DASG-CS effectively. (3) Propose a three-stage process integrating feature extraction, multi-source knowledge integration, and natural language generation. (4) Experiments on MS-COCO, Flickr30k, and Visual Genome with BLEU-n indices show that the method provides a richer description, especially in capturing spatial and depth relationships.

The rest of the article includes: Section 2 reviews related works, including approaches to object detection and spatial feature extraction, spatial relationships, Knowledge-Enhanced image captioning, and Natural Language generation for image captioning. Section 3 details the theoretical basis for proposed method. Section 4 analyzes the design and evaluation of experimental results on the MS-COCO [9] and Flickr30k [20] datasets. Section 5 discusses conclusions and development directions.

2 Related Work

The problem of image captioning has evolved significantly with the combination of visual models, linguistic models, and knowledge representation techniques. These works can be grouped into four main directions: (1) object recognition and extraction methods to form the basis for scene representation; (2) techniques for modeling scene relationships and structures, including scene graphs and 3D scene comprehension models; (3) knowledge enhancement approaches using Knowledge Graphs to supplement semantic and commonsense information; and (4) methods for generating natural language from structural or multimodal representations. This classification clarifies the progress of the field while highlighting the remaining limitations that our approach aims to address.

Traditional image captioning systems are primarily based on 2D visual features extracted from RGB images, which are the main input for the encoding-decoding architecture in image captioning [15, 17]. In the field of object recognition, the YOLO family has developed strongly from the YOLOv1 model with a single-stage detection architecture [12], then expanded through several improved generations such as YOLOv2, YOLOv3, and recent optimized variants [2, 13]. Modern versions such as YOLOv11 continue to improve inference accuracy and speed, especially in contexts with many small objects or dense scenes.

Although object detection models have made significant strides, much of the research still focuses on using 2D bounding boxes to represent object position. This representation fails to capture the three-dimensional nature of the scene, limiting the ability to infer depth, proximity, and the environment's layered structure. The lack of depth information makes it difficult for models to build a true spatial understanding, a crucial component for tasks that require complex inference of scene layout. Scene structure modeling plays a crucial role in the semantic understanding and spatial composition of images. One prominent approach is scene graph generation, where objects and the relationships between them are represented as clearly structured graphs [19, 22]. These scene graphs are often built through entity recognition and extraction of semantic relationships such as "person-sitting-on-chair" or "dog-standing-next-to-person", which enhances inference capabilities and improves many tasks such as captioning, image querying, and scene comprehension.

However, most current scene graph methods focus on semantic relationships and do not fully describe quantitative spatial relationships, such as distance, near or far degrees, or depth plane stratification. The lack of this precise spatial information means that scene graphs only partially reflect the nature of the scene and fail to represent the three-dimensional (3D) structure required by spatial inference.

To overcome this limitation, our approach incorporates depth estimation into the scene graph construction process. This allows for more informative modeling of spatial relationships, including semantic connections and quantitative representations related to the relative position and 3D structure of the scene. As a result, the generated scene graph is more accurately reflected in terms of spatial layout, thereby better supporting the task of generating spatially perceptive image descriptions.

Knowledge Graphs (KGs) play a crucial role in extending the semantic understanding of models by providing a clear knowledge structure, including entities, attributes, and relationships. In the field of computer vision, Visual Genome [6] is one of the largest visual knowledge sources, containing detailed annotations of objects, attributes, and relationships, forming the basis for building large scale visual knowledge graphs. Similarly, Flickr30k [20] provides tens of thousands of image-caption pairs, becoming a key dataset in training visual linguistic models.

Many previous studies have exploited external knowledge to enhance descriptive generation capabilities. Knowledge Graphs have been used to support visual inquiry answering (VQA) and image classification [10], demonstrating the effectiveness of adding background knowledge to visual models. Entity linking techniques help map detected objects in images to entities in the knowledge base, thereby enabling the system to leverage realworld knowledge. In addition, knowledge-based methods [16, 21] have demonstrated that integrating common sense and factual knowledge improves caption quality, especially in expressing action, meaning, and context. However, most current approaches only integrate knowledge at the caption generation stage, meaning the model only references knowledge when creating text, instead of using knowledge to structure the scene representation from the beginning. Unlike these approaches, our method builds a unified knowledge graph before the description generation stage, by combining visual observations from Visual Genome and Flickr30k with external knowledge sources. The integration of knowledge from multiple sources helps to form a more comprehensive scene representation, reflecting both image content and encompassing realworld understanding, thereby supporting a more semantically accurate and accurate caption generation model.

Natural language is the final component in the image description generation system. Early encoder-decoder models [5, 15] based on RNNs were used to generate sentences from visual features. The introduction of attention mechanisms [1, 17] helped the model focus on important image regions, thereby significantly improving description quality. Recently, Transformer-based architectures [3, 7] have further expanded the ability to model long-term context, becoming the dominant choice for captioning problems. In particular, the T5 model [11] with its "text-to-text" approach provides a unified processing framework for many language generation tasks.

In this study, we use text-to-text transformers to generate descriptions from structured semantic representations. By refining the model on data that clearly encodes spatial relationships and depth information, the system aims to create a description that accurately reflects the 3D spatial structure of the scene. Current paper on image captioning generation has made significant progress through object detection, scene graphs, knowledge integration, and Transformer-based linguistic models. However, most methods remain limited to two-dimensional visual representation, failing to fully model three-dimensional spatial relationships and depth information. Furthermore, commonsense knowledge is primarily exploited during the description generation phase, rather than being directly integrated into the structural representation of the scene. These limitations highlight the need for a unified scene representation model capable of combining depth information, spatial relationships, and semantic knowledge to improve image description quality.

3 Theoretical basis and proposed model

3.1 Theoretical Basis

The theoretical foundations used for image caption extraction methods are presented, including:

(1) Object Detection and Spatial Analysis Framework, Depth Estimation and Spatial Reasoning. In this study, YOLOv11 is used for object recognition, which is considered the first step in transforming raw visual information into structured components for relationship extraction, knowledge integration, and the generation of spatially perceptive image descriptions [13].

The Depth-Anything-V2 model [18] is incorporated to estimate depth information such as:

$$f_{depth} : \mathbb{R}^{H\times W\times 3} \to \mathbb{R}^{H\times W} \tag{1}$$

The depth map $D = f_{depth}(I)$ provides spatial context that enhances object relationship understanding. Combined with the Shapely geometry library [4], we compute spatial relationships between detected objects using geometric operations:

$$R_{spatial}(d_i, d_j) = \{distance(d_i, d_j), overlap(d_i, d_j), relative\_position(d_i, d_j)\} \tag{2}$$

(2) Knowledge Graph Construction:

Entity Linking and Knowledge Fusion: given the detection set $\mathcal{D}$ from Stage 1, we define the entity linking function as:

$$f_{link} : \mathcal{D} \times \mathcal{KB} \to \mathcal{E}_{linked} \tag{3}$$

where $\mathcal{KB}$ represents the external knowledge base and $\mathcal{E}_{linked}$ is the set of linked entities. The linking process employs semantic similarity scoring:

$$sim(d_i, e_k) = \alpha \cdot sim_{text}(c_i, label(e_k)) + \beta \cdot sim_{context}(context(d_i), context(e_k)) \tag{4}$$

where α and β are weighting parameters, and sim_{text} and $sim_{context}$ represent textual and contextual similarity measures respectively.

Knowledge Graph Formalization: The constructed knowledge graph is formally defined as a directed graph $KG = (V, E, R, A)$ where $V = \{v_1, v_2, ..., v_m\}$ is the set of entity vertices, $E \subseteq V \times V$ represents the edges between entities, R is the set of relation types, and $A : V \to \mathcal{P}(\mathcal{A})$ maps entities to their attribute sets.

(3) T5-based Natural Language generation framework: In this work, T5 is adopted as a large pretrained language model (LLM) responsible for structured knowledge grounding and caption generation, while visual and spatial reasoning are handled before the language modeling stage [11].

Text-to-Text Transfer Learning Our approach leverages the T5 (Text-to-Text Transfer Transformer) model [11] for generating natural language descriptions from structured semantic representations. The T5 framework treats all NLP tasks as text-to-text problems:

$$f_{T5} : \mathcal{S}_{semantic} \to \mathcal{T}_{text} \tag{5}$$

where $\mathcal{S}_{semantic}$ represents the semantic input derived from the knowledge graph and $\mathcal{T}_{text}$ is the generated natural language output [11].

Semantic Representation Encoding The knowledge graph entities and relationships are encoded into a structured semantic representation $S_{semantic}$ that serves as input to the T5 model:

$$S_{semantic} = encode(KG, R_{spatial}, C_{context}) \tag{6}$$

where $C_{context}$ represents the contextual information derived from the enriched data analysis [22].

Attention Mechanism for Semantic Focus The T5 model employs multi-head self-attention to focus on relevant semantic elements:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{7}$$

where Q, K, and V represent query, key, and value matrices respectively, derived from the semantic representation [14].

(4) Contextual reasoning and inference: This section presents a contextual inference mechanism based on data enriched from knowledge graphs. The proposed method does not require explicit modeling of relationships but directly utilizes extended semantic information to infer relevant context [16].

Enriched Data Analysis Our approach performs contextual reasoning using enriched data from the knowledge graph without explicit relationship modeling. The contextual inference function is defined as:

$$f_{inference} : \mathcal{E}_{enriched} \rightarrow \mathcal{I}_{context} \tag{8}$$

where $\mathcal{E}_{enriched}$ represents the enriched entity set and $\mathcal{I}_{context}$ denotes the inferred contextual information [16].

Semantic Coherence Optimization To ensure semantic coherence in the generated descriptions, we optimize the following objective function:

$$\mathcal{L}_{total} = \mathcal{L}_{generation} + \lambda_1 \mathcal{L}_{coherence} + \lambda_2 \mathcal{L}_{semantic} \tag{9}$$

where $\mathcal{L}_{generation}$ is the standard language modeling loss, $\mathcal{L}_{coherence}$ measures textual coherence, $\mathcal{L}_{semantic}$ ensures semantic consistency with the input knowledge graph, and λ_1, λ_2 are regularization parameters [11].

(5) Multi-Modal integration: The alignment function is used to map these three feature spaces into a unified representation space, facilitating consistent information merging. Multimodal merging is achieved through a weighted combination of features transformed by the respective transformation functions of each method, with weights optimized to reflect the relative importance of each information source.

Cross-Modal Alignment The integration of visual features, spatial information, and semantic knowledge requires cross-modal alignment. We define the alignment function as:

$$f_{align} : \mathcal{F}_{visual} \times \mathcal{F}_{spatial} \times \mathcal{F}_{semantic} \rightarrow \mathcal{F}_{unified} \tag{10}$$

where $\mathcal{F}_{visual}$, $\mathcal{F}_{spatial}$, and $\mathcal{F}_{semantic}$ represent visual, spatial, and semantic feature spaces respectively, and $\mathcal{F}_{unified}$ is the unified representation space [8].

Information Fusion Strategy The multi-modal information fusion employs weighted combination of feature representations:

$$\mathcal{F}_{unified} = \sum_{i=1}^{3} w_i \cdot \phi_i(\mathcal{F}_i) \tag{11}$$

where ϕ_i represents the transformation function for each modality, and w_i are learned weights that adapt to the relative importance of each information source [3].

3.2 Proposed method for generating image captioning

A proposed architecture (Fig. 1) consists of three main stages: (1) Spatial feature Extraction; (2) Knowledge graph construction; (3) Depth-Aware Caption Generation.

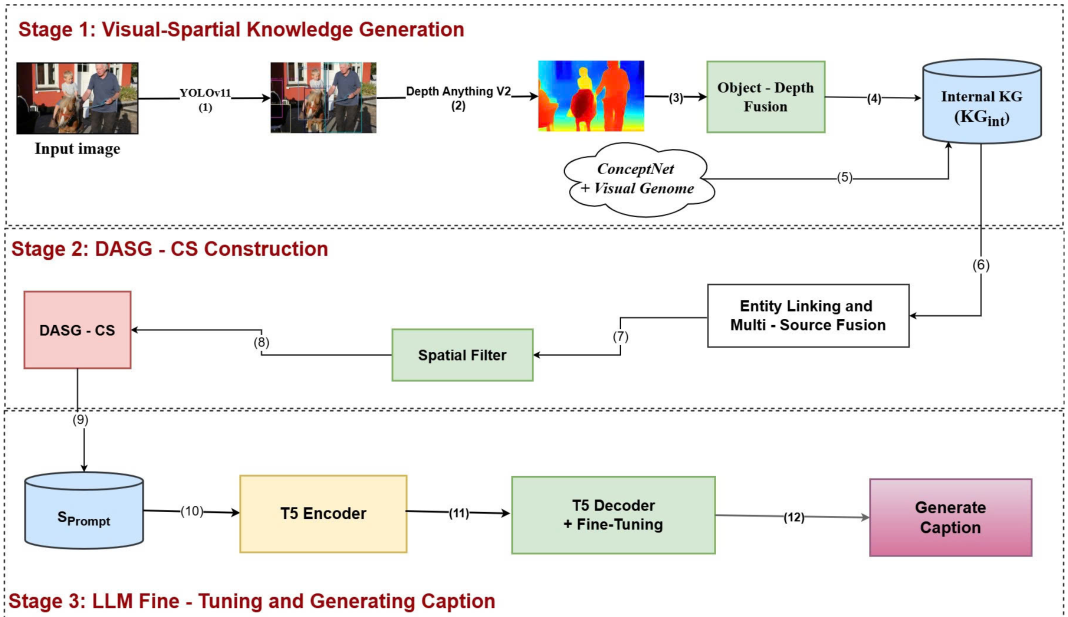

Fig. 1. Structure of proposed model for image captioning

3.2.1. Initial Visual-Spatial Knowledge Generation

The first stage of our system focuses on generating a Knowledge Graph (KG) that encodes both the semantic entities and their derived spatial-structural relationships within the input image. This initial graph, denoted as the Internal Knowledge Graph (KGInt), serves as the foundational, image-grounded structure before external knowledge enrichment.

Stage 1 encompasses the feature extraction, object-depth fusion, and the construction of the KGInt (Fig. 2).

Object and Depth Feature Extraction employs a dual-branch feature extraction mechanism.

Object Detection: The YOLOv11 model is used to detect objects and extract attributes. The output is a set of detections $D = \{d_i\}^n$, where each d_i includes the bounding box B_i, class label C_i, and confidence score S_i.

Depth Estimation: The Depth Anything V2 model generates a high-resolution, dense Depth Map $D_{map} = \mathbb{R}^{H\times W}$. This map provides the crucial relative depth information necessary for 3D spatial reasoning.

Object-Depth Fusion and Spatial Logic: The Fusion and Spatial Logic modules convert the raw visual and depth data into structured relationships.

1. **Quantification and Fusion:** The bounding boxes B_i are aligned with D_{map}. We compute the Average Relative Depth ($\bar{d}_i$) for each object O_i within its bounding box. This step yields a set of Rich Objects R, where each object entity is augmented with its spatial center and $\bar{d}_i$. The $\bar{d}_i$ values are used exclusively for internal calculation of relative depth and are not stored as absolute values in the final KG.

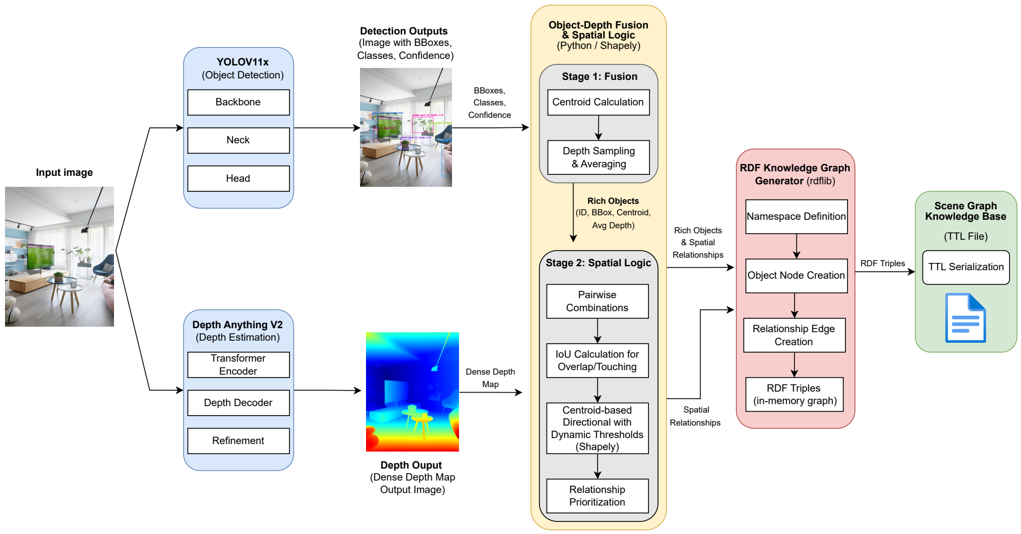

Fig. 2. An architecture of Initial Visual-Spatial Knowledge Generation

2. **Spatial Predicate Generation:** The core of this stage is the derivation of new spatial predicates. By comparing the difference in average depth $\mathbf{\Delta d_{i,j}} = \bar{d}_i - \bar{d}_j$ between object pairs (O_i, O_j), we generate a Novel Set of Qualitative 3D Spatial Predicates $(\mathcal{P}_{3D})$:

   $$\mathcal{P}_{3D} \subset \{\text{is\_closer\_than, is\_farther\_than, on\_same\_depth\_plane\_as, in\_immediate\_foreground, etc.}\}$$

   This logic uses dynamic thresholds to convert the continuous $\Delta d_{i,j}$ into discrete, relational triplets. Traditional 2D relationships (overlap, adjacency) are computed alongside these 3D predicates (using methods like IoU and Centroid-based Directional logic).

Internal Scene Graph Construction the resulting rich Oobjects and the generated 2D or 3D relationships are fed into the RDF Knowledge Graph Generator (utilizing libraries such as rdflib):

1. Namespace definition and object node creation (from R).
2. Relationship edge creation (from P3D and 2D predicates).

This process finalizes the Internal Knowledge Graph (KG_{Int}), a set of RDF Triples that accurately models the scene's visual and spatial configuration. This KG_{Int} is then serialized as the output of stage 1, ready for external knowledge enrichment in stage 2.

3.2.2. DASG-CS Construction and Multi-Source Fusion

The primary objective of stage 2 is the construction of the Depth and Commonsense Augmented Scene Graph (DASG-CS) by integrating the Internal Knowledge Graph (KGInt) with external knowledge resources. This stage addresses the limitation of purely visual KGs by introducing contextual reasoning capabilities (Fig. 3).

Knowledge source acquisition three distinct knowledge streams are prepared for fusion:

1. Internal visual and 3D spatial triples (KGInt) containing 2D semantic triples and a set of qualitative P3D spatial predicates.
2. Semantic triples (KGFact) incorporates standard semantic relationships retrieved from large-scale visual knowledge bases (Visual Genome and Flickr), providing general object-to-object semantic links.

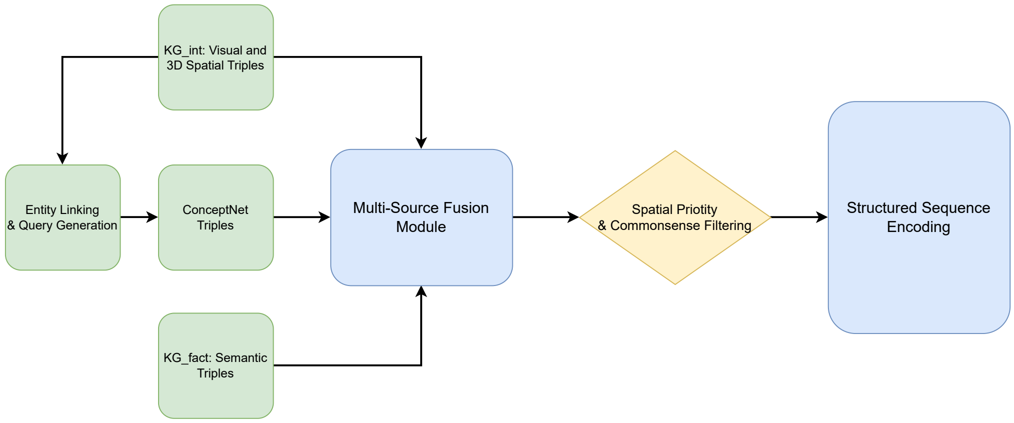

Fig. 3. Architecture of the DASG-CS Construction and Fusion Module

3. Commonsense triples (KGCS) is acquired through a two-step process: Entity linking and query generation that uses the object entities (Nodes) from KGInt as queries to retrieve relevant triples from the ConceptNet. This yields KGCS, which enriches the graph with a human-like understanding of context, purpose, and causality.

Multi-Source fusion and graph consolidation the three knowledge streams (KGInt, KGFact, KGCS) converge into the Multi-Source fusion module.

1. Entity alignment: The module first ensures all entities (subjects and objects) are correctly aligned and unified across the three sources.
2. Consolidation: The module performs a set union operation to form the raw DASG-CS: DASG-CS = $KG_{Int} \cup KG_{Fact} \cup KG_{CS}$
3. During consolidation, KGInt (the direct visual evidence) is assigned the highest priority to resolve any potential conflicts or inconsistencies arising from general/abstract knowledge found in KG_{Fact} or KG_{CS}.

Spatial priority and commonsense filtering: The raw DASG-CS (Dense Annotated Scene Graph for Commonsense) typically contains redundant scene information and is high-dimensional. To address this, we introduce a filtering mechanism (represented by the diamond shape in Fig. 3) to retain only the most impactful triples (subject-predicate-object relationships) for caption generation: (1) Spatial priority filtering: Triples containing the novel 3D spatial predicates (P3D, which refer to relationships describing 3D spatial arrangements) are automatically prioritized and flagged. This ensures that the large language model's (LLM) final output is strongly grounded in the scene's structural awareness; (2) Commonsense filtering: KGCS (Knowledge Graph for Commonsense) triples are filtered based on a computed relevance score by measuring semantic similarity between the triple and the image context, thereby removing weak or generic commonsense links that might introduce noise.

This filtering step yields a set of high-quality, non-redundant, prioritized triples, which Topt uses for the final encoding. The final step of stage 2 is structured sequence encoding. Here, the optimized set of triples (Topt) is serialized into a single input sequence, S_{prompt}, for the LLM. This encoding uses specific delimiter tokens (`[3D_START]`, `[CS_FACT]`) to clearly distinguish the type of knowledge, allowing the LLM in Stage 3 to explicitly learn to weight and utilize each semantic and spatial component effectively.

3.2.3. Spatially-Aware LLM Fine-Tuning

The final stage uses the structured representation Sprompt to fine-tune a language model capable of producing spatially accurate and contextually rich image captions. This stage transforms multi-level knowledge (3D, 2D, commonsense, factual) into natural and coherent descriptions (Fig. 4).

Stage 3: Spatially-Aware LLM Fine-Tuning

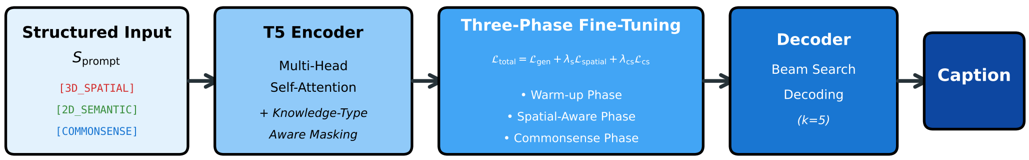

Fig. 4. Architecture of the Spatially-Aware LLM Fine-Tuning

Model selection and architecture: In this paper, the T5 model is used for unified text-to-text design and strong generalization across NLP tasks. Caption generation is treated as a sequence-to-sequence translation process in which the model maps the structured input into a natural caption.

Input representation and tokenization: The sequence Sprompt is organized using explicit delimiters that separate knowledge categories and help the model attend to them correctly. This structure is formatted to guide the model to emphasize spatial predicates for positional descriptions and commonsense facts for functionality or behavior, such as:

[3D_START] car is_closer_than building [3D_END]
[2D_START] person on sidewalk [2D_END]
[CS_START] car UsedFor transportation [CS_END]
[FACT_START] building IsA structure [FACT_END]

Training objective: The model is optimized using a combination of three components: The model is optimized using a combination of three objectives: (i) a caption generation objective that ensures fluent and coherent output aligned with ground-truth captions, (ii) a spatial consistency objective that encourages the model to produce spatial expressions matching the 3D relations encoded in the input, and (iii) a commonsense grounding objective that increases the semantic alignment between generated descriptions and the commonsense triples provided from stage 2. Two weighting coefficients control the importance of spatial alignment and commonsense grounding during training.

Fine-Tuning strategy: The fine-tuning proceeds in three phases: (1) warm-up phase: train on standard captioning datasets (MS-COCO) to retain linguistic fluency; (2) spatial-aware phase: introduce spatial consistency supervision and gradually increase its weight so the model learns to use accurate positional terms; (3) commonsense phase: activate the commonsense objective to integrate functional and causal knowledge into the generated captions.

4 Experimental and evaluation results

4.1 Environment and experimental data

The experiment is built on the Python Notebook platform. The graphs are built based on the Python Matplotlib library. The process of the experiment is built on the Dell Gaming G3 15 computer environment, which includes an Intel(R) Core(TM) i7-10750H processor, 16GB RAM, GTX 1660 ti GPU, and Windows 11 operating system. The experimental data sets are MS-COCO and Flickr30k multi-object image sets, which are described in Table 1.

4.2 Experiment and evaluation of results

The experimental results of building captions for images on the MS-COCO and Flickr30k datasets are shown in Figs. 5 and 6 (`IC_DSGLM`). The figure 5 shows the results of of genereated Scene Graph and

Table 1. Detailed description for the ViVQA and OpenViVQA datasets

Datasets	Images in Validation	Concept in the Visual Concept
MS-COCO [9]	5,000	80
Flickr30k [20]	1,000	79

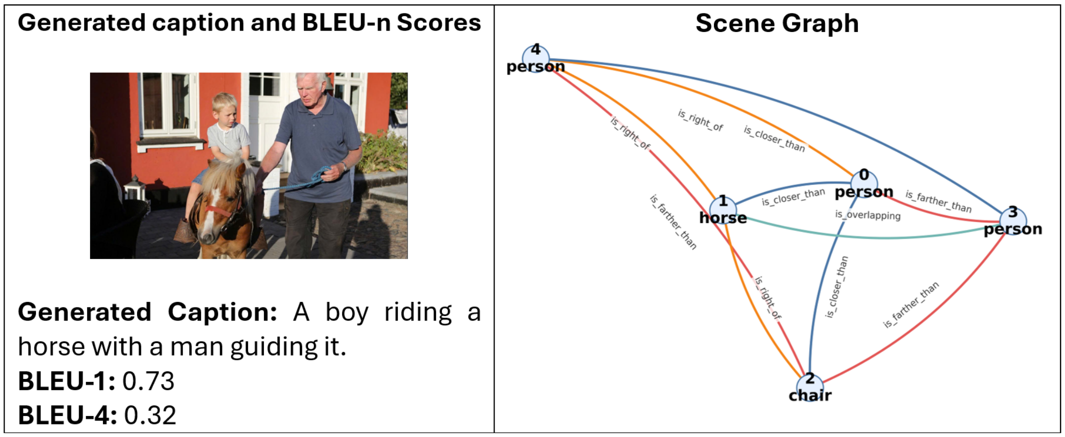

Fig. 5. illustrating the results of Scene Graph and building captions image 000000050380.jpg (MS-COCO)

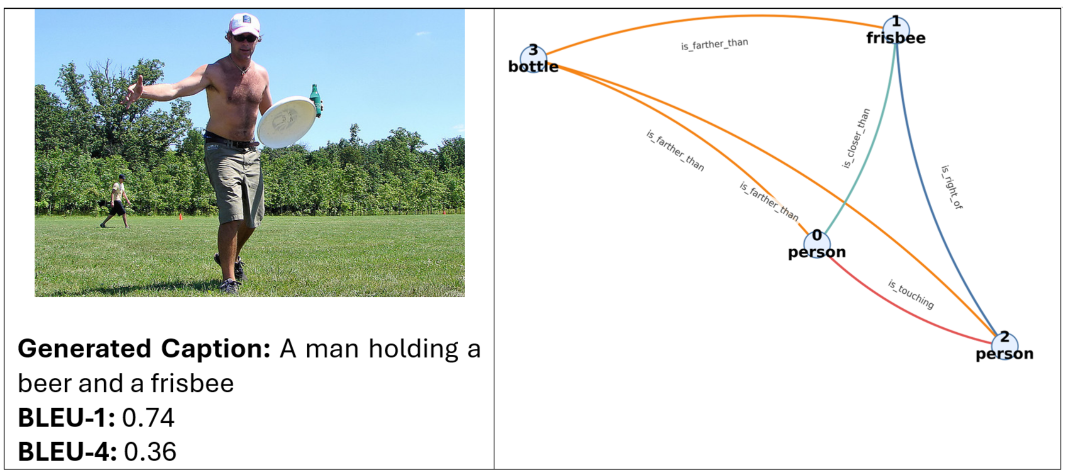

Fig. 6. illustrating the results of Scene Graph and building captions image 000000025057.jpg (Flickr30k)

Table 2. Experimental results of image captioning using BLEU-1 and BLEU-4

Datasets	Number of validation images	BLEU-1	BLEU-4
MS-COCO [9]	5,000	0.7339	0.3110
Flickr30k [20]	1,000	0.7686	0.3428

building captions for the image 000000050380.jpg (`MS-COCO`). Figure 6 shows the results of genereated Scene Graph and building captions for the image 000000025057.jpg (`Flickr30k`). The average accuracy results of these two image sets are presented in Table 2 with BLEU-1 and BLEU-4 metrics.

Table 2 shows that the proposed method achieves high performance on both the MS-COCO and Flickr30k datasets. Specifically, it obtains 0.7239 (BLEU-1), 0.3110 (BLEU-4), and 0.7686 (BLEU-1), 0.3428 (BLEU-4), respectively. These results indicate that integrating depth-aware spatial relationships and multi-source knowledge graphs enhances caption generation quality. This is particularly evident in capturing fine-grained semantic and structural information in the images. Table 3 further contextualizes these contributions. It presents a comparative analysis with recent methods that report lower performance under similar evaluation settings. This comparison highlights the effectiveness of our depth-aware and knowledge-enhanced modeling strategy.

Table 3. Comparing image caption accuracy with some other methods on MS-COCO

Method	**BLEU-1**
Bottom-Up Attention [1]	0.7700
M2 Transformer [3]	0.7760
Visual Relationship [19]	0.7350
IC_DSGLM	**0.7639**

Table 3 presents a comparative evaluation of the proposed IC_DSGLM with recent image captioning methods on the MS-COCO. Table 4 reports the corresponding results on Flickr30k. Although training settings may vary across methods, all comparisons are conducted on the same datasets and reported using standard BLEU-n metrics, providing a reasonable and commonly adopted evaluation protocol for image captioning.

Table 4. Comparing image caption accuracy with some other methods on Flickr30k

Method	**BLEU-1**	**BLEU-4**
Bottom-Up Attention [1]	0.7100	0.2700
M2 Transformer [3]	0.7550	0.3100
Visual Relationship [19]	0.7200	0.2850
IC_DSGLM	**0.7686**	**0.3428**

Figure 7 shows the training and validation loss curves during the fine-tuning process of the proposed model. Both losses consistently decrease across epochs, indicating stable convergence. The small gap between training and validation loss suggests good generalization and no significant overfitting. These results confirm the effectiveness of the spatially-aware fine-tuning strategy.

The performance of IC_DSGLM compared to the methods reported in Tables 3 and 4 can be attributed to several factors: First, prior approaches that rely primarily on 2D visual features or attention-based mechanisms, IC_DSGLM explicitly incorporates depth-aware spatial relationships, enabling more accurate modeling of object proximity, relative position, and scene structure. Second, the proposed method constructs a unified depth and commonsense augmented Scene Graph, allowing multi-source semantic and commonsense knowledge to be integrated at the representation level rather than only at the language generation stage. Finally, the spatially-aware LLM fine-tuning strategy enables the language model to effectively leverage structured spatial and semantic cues, resulting in more precise and contextually rich captions.

5 Conclusion and development

This paper introduces IC_DSGLM, a tool that creates image captions using information about depth, connections between things in the image, and knowledge from other sources. By building special scene graphs that add both depth and basic world knowledge, this method helps describe where things are in the scene and what they mean. Tests on two popular datasets, MS-COCO and Flickr30k, show that

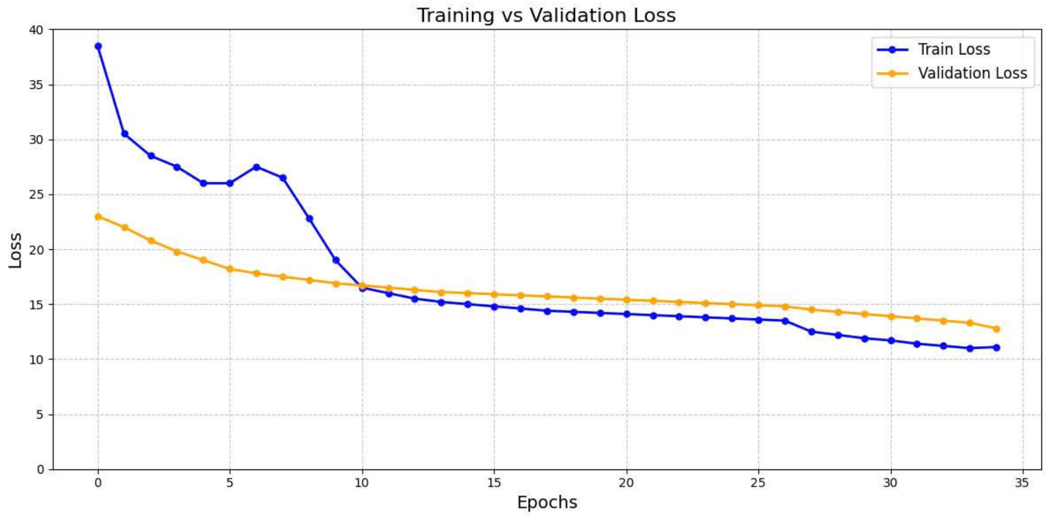

Fig. 7. Training and validation loss curves during the fine-tuning of the proposed spatially-aware image captioning model

IC_DSGLM gives more accurate and clear captions, especially about positions, depth, and the overall layout of the scene. However, the current model relies on monocular depth estimation, which may introduce noise in complex scenes, and the evaluation is limited to two benchmark datasets. Future work will focus on extending the framework with more robust depth estimation techniques, incorporating larger-scale vision language models such as BLIP-2 or LLaVA, and exploring more advanced reasoning mechanisms over knowledge graphs to further enhance spatial inference and semantic consistency in image caption generation.

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Disclosure of Interests

The authors have no competing interests to declare that are relevant to the content of this article.

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